

Notes on Dzanic, Huanga From & Sauret Article

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NB: Notes for Chapter 2.1, Chapter 2.4, Chapter 3.1-3.1.1 and Chapter 3.2-3.2.3 from Dzanic, Huanga From & Sauret's Lattice Boltzmann methods for simulating non-Newtonian fluids: A comprehensive review. Anything in bold purple has a footnote with a question.

The lattice Boltzmann method

LBM for simple hydrodynamics

Contrasting the continuum based NSE, the kinetic theory LBM is a description of particle groups using distribution functions f_α , weights w_α and lattice velocities ξ_α and we have $c_s^2 \mathbf{I} = \sum_\alpha w_\alpha \xi_\alpha \xi_\alpha$ with c_s the lattice dependent speed of sound.

Using D_2Q_9 , we have the velocity set

$$\xi_\alpha = \begin{cases} (0, 0) & \alpha = 0 \\ (1, 0), (-1, 0), (0, 1), (0, -1) & \alpha = 1, 2, 3, 4 \\ (1, 1), (-1, 1), (-1, -1), (1, -1) & \alpha = 5, 6, 7, 8 \end{cases} \quad (1)$$

the lattice weights

$$w_\alpha = \begin{cases} 4/9 & \alpha = 0 \\ 1/9 & \alpha = 1, 2, 3, 4 \\ 1/36 & \alpha = 5, 6, 7, 8 \end{cases} \quad (2)$$

The proportion of particles moving with α^{th} lattice velocity at site $\mathbf{x}$ at time t is given by

$$f_\alpha(\mathbf{x} + \xi_\alpha \Delta t, t + \Delta t) = f_\alpha(\mathbf{x}, t) + \Omega_\alpha^{\text{SRT}}(\mathbf{x}, t) \quad (3)$$

where $f_\alpha(\mathbf{x} + \xi_\alpha \Delta t, t + \Delta t)$ is the streaming step and the collision operator

$$\Omega_\alpha^{\text{SRT}}(\mathbf{x}, t) = -\frac{\Delta t}{\tau} [f_\alpha(\mathbf{x}, t) - f_\alpha^{\text{eq}}(\mathbf{x}, t)] \quad (4)$$

is the single relaxation time using the BGK approximation and relaxation time τ is related to the kinematic viscosity ν by $\nu = c_s^2(\tau - \frac{1}{2})\frac{\Delta x^2}{\Delta t}$ typically with $\Delta x = \Delta t = 1$ in lattice units.

The equilibrium term $f_\alpha^{\text{eq}}(\mathbf{x}, t)$ is given by the Maxwell-Boltzmann distribution approximated by Hermite polynomials to second order,

$$f_\alpha^{\text{eq}}(\mathbf{x}, t) = w_\alpha \rho \left\{ 1 + \frac{\boldsymbol{\xi}_\alpha \cdot \mathbf{u}^{\text{eq}}}{c_s^2} + \frac{1}{2} \left[\left(\frac{\boldsymbol{\xi}_\alpha \cdot \mathbf{u}^{\text{eq}}}{c_s^2} \right)^2 - \left(\frac{u^{\text{eq}}}{c_s} \right)^2 \right] \right\} \quad (5)$$

where $\mathbf{u}^{\text{eq}}$ is the equilibrium velocity. The macroscopic variables can also be recovered from the moments of $f_\alpha(\mathbf{x}, t)$,

$$\rho = \sum_\alpha f_\alpha \quad (6)$$

$$\rho \mathbf{u} = \sum_\alpha f_\alpha \boldsymbol{\xi}_\alpha \quad (7)$$

The link back to NSE can be found through analysis, resulting in

$$\boldsymbol{\nabla} \cdot \mathbf{u} = 0, \quad \frac{D\mathbf{u}}{Dt} = -\frac{1}{\rho} \boldsymbol{\nabla} p + \nu \boldsymbol{\nabla}^2 \mathbf{u} \quad (8)$$

The method can then be summarised as

1. Collision step

$$f_\alpha^*(\mathbf{x}, t) = f_\alpha(\mathbf{x}, t) + \Omega_\alpha^{\text{SRT}}(\mathbf{x}, t).$$

2. Streaming step

$$f_\alpha(\mathbf{x} + \boldsymbol{\xi}_\alpha \Delta t, t + \Delta t) = f_\alpha^*(\mathbf{x}, t).$$

3. Macroscopic properties using moments

$$\rho(\mathbf{x}, t) = \sum_\alpha f_\alpha(\mathbf{x}, t)$$

$$\mathbf{u}(\mathbf{x}, t) = \frac{1}{\rho(\mathbf{x}, t)} \sum_\alpha f_\alpha(\mathbf{x}, t) \boldsymbol{\xi}_\alpha$$

Additional momentum contributions in LBM

Adding forces to LBM can be achieved by introducing a forcing term $\Omega_\alpha^F(\mathbf{x}, t)$ during the collision step,

$$f_\alpha(\mathbf{x} + \boldsymbol{\xi}_\alpha \Delta t, t + \Delta t) = f_\alpha(\mathbf{x}, t) + \Omega_\alpha^{\text{SRT}}(\mathbf{x}, t) + \Omega_\alpha^F(\mathbf{x}, t) \quad (13)$$

The forms of this forcing term vary, here are two. The explicit forcing (EF),

$$\Omega_\alpha^F(\mathbf{x}, t) = \Delta t \left(1 - \frac{1}{2\tau} \right) \frac{\mathbf{F} \cdot (\boldsymbol{\xi}_\alpha - \mathbf{u}^{\text{eq}})}{\rho c_s^2} f_\alpha^{\text{eq}} \quad (14)$$

where the equilibrium velocity $\mathbf{u}^{\text{eq}} = \sum_{\alpha} \boldsymbol{\xi}_{\alpha} + \frac{\Delta t}{2\rho} \mathbf{F}$, is used to compute f_{α}^{eq} in (5).

Similarly, the forcing scheme proposed by Guo et al has the form,

$$\Omega_{\alpha}^F(\mathbf{x}, t) = \Delta t \left(1 - \frac{1}{2\tau} \right) w_{\alpha} \left(\frac{1}{c_s^2} (\boldsymbol{\xi}_{\alpha} - \mathbf{u}^{\text{eq}}) + \frac{1}{c_s^4} \xi_{\alpha}^2 \mathbf{u}^{\text{eq}} \right) \cdot \mathbf{F} \quad (15)$$

with $\xi_{\alpha}^2 = |\boldsymbol{\xi}_{\alpha}|^2$ and the same equilibrium velocity $\mathbf{u}^{\text{eq}} = \sum_{\alpha} \boldsymbol{\xi}_{\alpha} + \frac{\Delta t}{2\rho} \mathbf{F}$, is used to compute f_{α}^{eq} in (5).

The addition of this forcing term produces,

$$\rho = \sum_{\alpha} f_{\alpha} \quad (16)$$

$$\rho \mathbf{u} = \sum_{\alpha} f_{\alpha} \boldsymbol{\xi}_{\alpha} + \frac{\Delta t}{2\rho} \mathbf{F} \quad (17)$$